

AIYSHWARIYA (Riya) Paulvannan Kanmani

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PROFILE

Experienced Applied Scientist with extensive expertise in statistical machine learning applications and research interest in environmental sustainability. Proficient in Python, PySpark, R, SQL, C/C++, MS Excel, and cloud and data platforms including Azure, Databricks, and Tableau. Specializes in machine learning, natural language processing (NLP), recommender systems, clustering, predictive modeling, data analytics & visualization, operations research, and big data.

EDUCATION

- Texas A&M University (TAMU)**, College Station, TX, USA May'26 – Present
Doctor of Philosophy, AI & Data Science in Civil Engineering
- Purdue University (PU)**, West Lafayette, IN, USA Aug'17 – July'19
Master of Science, Industrial Engineering
- **Thesis:** *Assessing Global Environmental Sustainability Via an Unsupervised Clustering Framework*, **Advisor:** Dr. Roshi Nateghi
- Indian Institute of Technology Madras (IIT M)**, India Aug'13 – July'17
Bachelor of Technology | Minor: Operations Research
- **Thesis:** *Electrodeposition of Copper Foam and Testing its Application in Denitrification*, **Advisor:** Dr. Lakshman Neelakantan

PROFESSIONAL EXPERIENCE

- Amazon | Applied Scientist** July'22 – May'26
Voice of Customer – Discovery Science
- Implemented an anomaly detection system to proactively identify operational defects impacting customer service
 - Expanded AI customer service chatbot feature to support semantic search-based anecdote retrieval
- Devices Demand Science:** *Built & deployed daily demand forecast ML models for all Blink devices for automated planning*
- Led the creation and deployment of daily demand forecast ML models for Blink devices, automated planning processes.
 - Drove scientific improvements across device families, leveraging dynamic lag and cross-cannibalization features.
- American Airlines - Operations Research & Advanced Analytics Group | Senior Data Scientist** Oct'19 – July'22
Developed & implemented AI solutions, including a maintenance fix recommender system and prediction models for risky aircrafts
- Engineered a maintenance fix recommender system using NLP on log texts, decreased re-work by 65%.
 - Developed prediction models targeting and prioritizing risky aircrafts, optimizing maintenance operations.
 - Investigated & rectified misdiagnoses in the engine part removal process, improving operational efficiency by 10%.
 - Implemented an ML solution based on propensity prediction, increasing customer email offer CTR by 40%.
 - Personalized ancillary purchase offers to traveling customers through machine learning propensity models.
- Purdue University | Graduate Teaching Assistant** Aug'18 – May'19
- Courses: Statistical Quality Control, Probability & Statistics, Optimization for Big Data (total 300+ students)
 - Created video tutorials for application of theoretical concepts using tools such as R to solve industry problems
- Franklin Templeton Investments | Operations Data Analyst Intern** Summer'16
- Identified critical areas for market capitalization through comprehensive exploratory data analysis.
 - Delivered recommendations resulting in a 40% increase in profit margin, fostering improved relations with the firm.

RESEARCH EXPERIENCE & PUBLICATION*

- *Assessing Global Environmental Sustainability Via an Unsupervised Clustering Framework** Aug'18 – July'19
- Proposed a machine learning framework to objectively assess global sustainability for environmental policy makers.
 - Utilized unsupervised learning theory to assess and track the sustainability of countries over a 10-year period.
 - Identified country-specific trajectories towards enhancing environmental sustainability.
- Multivariate prediction of Greenhouse-gas Emissions** Spring'18
- Developed & implemented a multivariate boosting model to predict 4 types of greenhouse-gas emissions.
 - Revealed underlying infrastructure of dependency between emissions & aspects of economy.

Reduction of Taxi Time in Air Traffic Control Systems: Simulation Modeling

Fall'17

- Created a model simulating standard international airport operations to detect taxi time influencing factors.
- Evaluated factors' effects through statistical regression and recommended optimization strategies.

Carpooling Routing Algorithms for Fuel Efficiency

Spring'16

- Simulated a Java algorithm predicting optimized fuel-efficient routes for carpooling on a real-life sub-network.
- Forecasted optimal routes with the highest fuel efficiency using typical travel demand information.

LEADERSHIP AND EXTRACURRICULARS

Amazon, Mentoring Program: Mentorship to interns for goal setting and career progression

American Airlines, University Relations (led Purdue hiring) **and Knowledge Sharing** (cloud platforms onboarding)

IIT Madras Aquatics: Represented the University and secured positions in inter-university sports meets

Head - Metallurgical & Material Student Association, IIT Madras: Elected in-charge and successfully executed all technical & cultural activities for the department